JAYA KRISHNA JAMMU

MSc Electrical Engineering · Junior Robotics Engineer

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TECHNICAL SKILLS

- Arduino / Embedded C/C++
- Robotics & Automation
- Sensor & Actuator Integration
- Electronics & Circuit Design
- Hardware Testing & Validation
- MATLAB / Simulink · Python
- Basic PLC Programming
- Data Acquisition
- Troubleshooting & Debugging

CERTIFICATIONS

Introduction to IoT
IIT Kharagpur (NPTEL)

Industrial Automation (PLC)
SV Technologies, 2024

WORKSHOPS

- Electrical Systems using MATLAB
- Industrial Automation using PLC
- Designing of Solar PV Systems

HONOURS & AWARDS

- **1st** Web Dev Hackathon, 42 Learn
- **1st** Technical Quiz, Rise Radiance, Rise Engineering College
- **2nd** Circuit Debugging, Bectagon, Bapatla Engineering College
- **2nd** Paper Presentation, Technopulse, RSR Engineering College

LANGUAGES

English Advanced (IELTS)
German Elementary (A1)
Hindi Intermediate
Telugu Native

PROFESSIONAL PROFILE

MSc Electrical Engineering student at FAU Erlangen-Nürnberg with hands-on industry experience as the first Junior Robotics Engineer at Koding Caravan, working under a Senior Robotics Engineer across the full development cycle — embedded C/C++, sensor integration, technical documentation, and hardware validation. Adapts quickly to new tools and workflows in fast-paced environments and consistently delivers reliable results. Seeking a working student or part-time role in embedded systems, robotics, or automation.

PROFESSIONAL EXPERIENCE

Junior Robotics Engineer

Koding Caravan · Sep 2025 – Mar 2026 · Singarayakonda, India · First junior hire, under Senior Robotics Engineer

- Worked under a Senior Robotics Engineer to help establish the company's robotics development capability from the ground up.
- Designed and built Arduino-based prototypes in C/C++, taking systems from concept through to validated hardware.
- Integrated sensors (ultrasonic, IR, IMU) and actuators via I2C/UART for real-time automation applications.
- Diagnosed hardware faults and software bugs under time pressure, improving system reliability, reducing downtime.
- Collaborated with founders to translate requirements into working prototypes within tight deadlines.
- Authored technical documentation, wiring schematics, and test reports to support ongoing product development.

EDUCATION

MSc Electromobility

Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU) · Apr 2026 – Present · Erlangen, Germany

B.Tech – Electrical & Electronics Eng.

Rise Krishna Sai Prakasam Group of Institutions (JNTUK) · Nov 2021 – Apr 2025 · Ongole, India

PROJECTS & PUBLICATIONS

Integrated Charging System for EV Power & Auxiliary Batteries Using Dual Active Bridge Converter

Designed and simulated a DC-DC Dual Active Bridge (DAB) converter to charge high-voltage traction and low-voltage auxiliary EV batteries from a single power stage, using closed-loop phase-shift modulation for bidirectional power flow and stable voltage regulation. Validated in MATLAB/Simulink across efficiency, transient response, and thermal behaviour; demonstrated elimination of separate onboard chargers, reducing EV system weight and cost.

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